Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/23/2017
PI Name:	Marr, David W.M. Ph.D.		
Project Title:	Fabrication, Crystallization, and Folding of Complex Colloidal Molecules under the Influence of Applied External Fields		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dmarr@mines.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-273-3720
Organization Name:	Colorado School of Mines		
PI Address 1:	Chemical and Biological Engineering		
PI Address 2:	1500 Illinois Street		
PI Web Page:			
City:	Golden	**State:**	CO
Zip Code:	80401-1887	**Congressional District:**	7
Comments:	https://		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 8/31/2019 per NSSC information (Ed., 10/2/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Ning (Colorado School of Mines) Solomon, Michael (University of Michigan, Ann Arbor)		
Grant/Contract No.:	NNX13AQ54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Colloidal particles are technologically important materials with applications ranging from consumer products to plasmonic metamaterials to drug delivery agents. With phase behaviors and assembly characteristics that closely resemble atoms and molecules, they are also excellent molecular analogues that can grow into functional materials with high efficiency. The range of colloidal structures and molecular mimics we can obtain however is currently limited and we remain far from truly accessing the diversity of phases desired for both deeper fundamental understanding and associated technological application. This is due primarily to limited control of the intrinsic colloidal interactions in conventionally isotropic building blocks and a reliance on non-directed assembly. To address these issues, we propose the use of anisotropic external fields including optical and hydrodynamic fields to create anisotropic colloidal building blocks (primarily on ground). We also propose the use of anisotropic electric fields to direct their assembly into complex structures (primarily in space). This combination of rational design of colloidal composition and architecture and versatile control of the range, strength, and directionality of interparticle interactions will allow the study of two outstanding scientific questions particularly appropriate for investigation at the International Space Station: (i) How does complex symmetry evolve during the crystallization of anisotropic colloidal molecules, such as achiral dimer particles, racemic mixtures of chiral tetramers, and pure enantiomeric tetramers? (ii) How will the collective fluctuations of nodes (tetramers) and linkers (colloidal chains) induce the formation of mixed networks with non-linear mechanical properties? Results from these investigations will lead to better control of the non-covalent assembly of molecules and scalable processing of advanced materials with tailored structures and properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In-situ manipulation of anisotropic interactions and dynamic pathways, based on rational colloidal particle design and proper use of external fields, could lead to crystalline and aperiodic structures beyond those seen in nature. With the wide variety of attributes that characterize anisotropic interactions, we will probe previously inaccessible regions of complex phase space. Our studies will move significantly beyond hard spheres and towards experimental models appropriate for studying fundamental questions associated with complex symmetries. For example, how does chirality of the assembled structure evolve during the nucleation and crystallization of achiral molecules, chiral molecules in racemic mixtures, or in pure enantiomers? Furthermore, our models of giant colloidal chains will allow for the manipulation and observation of the effect of site-specific interactions on chain dynamics at experimentally accessible length and time scales. Models that can be easily visualized like these will eventually lead to better understanding of complex phenomena such as polymer chain dynamics, chain reptation, interpenetrating double network dynamics, and protein folding. Results from this study will lead to improved control of non-covalent assembly of molecules, efficient tailoring of lattice symmetries, and the scalable processing of nano-structured materials. The development of new colloidal molecules and their associated assembled structures may have considerable technological impact as well. Anisotropic interactions developed in this proposal, both at the particle level and via applied fields, have been shown generally to lead to arrays with reduced symmetry and enhanced directionality. Such structures interact with a broad range of electromagnetic radiation in unique ways and can exhibit collective photonic, plasmonic, mechanical, electronic, or magnetic properties that are not manifested at the level of single particles. As a result, they have significant potential as next-generation functional materials. For example, dielectric arrays with diamond-like or quasicrystalline lattices have been proposed as ideal photonic crystals that can manipulate light propagation in three dimensions, forming the basis for next-generation all-optical communication technologies. Chiral tetramers studied here could exhibit distinct plasmonic properties resulting from the collective coupling between particles in the pyramid, as confirmed in numerical modeling, which have great potential for electromagnetic metamaterials and high-resolution sensors. The understanding of mirror-symmetry breaking in racemates is also important for developing improved strategies for separation of pharmaceutical molecules. In addition, anisotropic particles can be used to generate new kinds of colloidal networks with well-defined coordination and connectivity. By combining families of tetramers and chains, double networks – two distinct, but interpenetrating, networks of nodes and linkers – could be formed with combined high mechanical stiffness and high toughness. Colloidal networks are very common in a range of pharmaceutical and consumer production applications because of their ability to impart a yield stress that can stabilize active components uniformly throughout the formulation. Developing methods to produce colloidal networks with defined coordination, connectivity, and topology would expand the use and performance of colloidal stabilizing networks in these materials. Ultimately, successful implementation of our proposed research will provide the fundamental knowledge necessary for the development of technologies to design and control matter with tailored structures and properties.
Task Progress:	1) We have performed ground based studies of the assembly and dynamics of colloidal chains and demonstrated that superparamagnetic microlassos can achieve reversible cargo capture, transport, and release under a 3D AC magnetic field. We have first shown that lassos can be fabricated by aligning magnetic beads into linear chains under a 1D DC magnetic field and locked in place. With our previous studies, we are able to vary flexible rigidity (defined as EI) EI nd note that more flexible chains with a smaller EI bend more easily and have a smaller separation between aligned arms (in 1D fields). 2) We have also measured and modeled the formation and size of loops that form upon application of a magnetic field. Here we have observed that, with application of rotating instead of static magnetic fields, loops in the chains form which subsequently rotate due to a net torque on the induced dipoles. The deformation of a flexible chain into a lasso can be understood as the result of minimization of the summed elastic and magnetic energies in which both chain flexural rigidity EI and field strength B0 are governing parameters. Therefore, the lasso structure we observe results from a balance, chains coil in a tight ring to have a long zippered tail but do so with a penalty from the bending energy. Our modeling indicates that the radius of a stable lasso R* scales with the -2/3 power of the field strength and the 1/3 power of the flexural rigidity. 3) We have studied and modeled the translation of chain loops upon application of a rotating magnetic field. If loops spin in the bulk fluid or in the plane of an associated surface, no translation occurs; however, if the spinning plane is canted relative to the surface, loops travel. Our modeling suggests the average lasso propulsion speed V is proportional to the product of its angular rotation rate omega (angular velocity), lasso radius R*, and cosine of camber angle theta c. To verify this, instantaneous velocity profiles were measured and correlated with loop anisotropies. 4) While the rotation of the loops leads to well controlled translation, loop structure provides the opportunity for cargo loading. To demonstrate this, we have investigated the use of chain loops as lassos and demonstrated their potential for translation and delivery of microcargo. A particular advantage of this approach is that both cargo loading and unloading

	are not based on chemical bonding making the method reversible solely via manipulation of the applied field. To capture cargo, a 2D rotating field B0 is gradually increased making a chain curl to tighten around a target. Since the lasso radius can be tuned via field strength, a wide size range of cargos can be loaded by a given lasso. 5) We explored concepts of lock-and-key binding in anisotropic systems by studying the pair binding of triangular prism particles at an interface. When located at an air-water interface these microfabricated particles undergo a distribution of binding trajectories that lead to tip-to-tip bonds and edge-to-edge bonds, among others. This binding generates open networks at the interface; by producing the binding with triangular prisms, we successfully introduced a set of nodal points into the system; these nodes can support the self-assembly of large-scale reconfigurable networks with useful mechanical properties.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Yang T, Tasci O, Neeves KB, Wu N, Marr DWM. "Magnetic Lassos for Reversible Microcargo Transport." Presented at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Solomon M, Ferrar J, Bedi DS, Zhou S, Mao X. "Self-assembly of thin, triangular prisms into open networks at a flat air-water interface." Annual March Meeting of the American Physical Society, New Orleans, LA, March 13-17, 2017. Bulletin of the American Physical Society. 2017 Mar;2(4): P17.00012. http://meetings.aps.org/Meeting/MAR17/Session/P17.12 ; accessed 8/3/17. , Mar-2017
Abstracts for Journals and Proceedings	Yang T, Tasci O, Neeves KB, Wu N, Marr DWM. "Magnetic Microlassos for Reversible Cargo Capture, Transport and Release." Presented at 91st Colloid and Surface Science Symposium, City College of New York, NY, July 9-12, 2017. 91st Colloid and Surface Science Symposium, City College of New York, NY, July 9-12, 2017. , Jul-2017
Articles in Peer-reviewed Journals	Yang T, Tasci TO, Neeves KB, Wu N, Marr DWM. "Magnetic microlassos for reversible cargo capture, transport, and release." Langmuir. 2017 Jun 13;33(23):5932-7. Epub 2017 Mar 27. https://doi.org/10.1021/acs.langmuir.7b00357 ; PubMed PMID: 28318267 , Jun-2017
Journal/Magazine covers	Yang T, Tasci TO, Neeves KB, Wu N, Marr DWM. "Cover image in Langmuir for article, 'Magnetic microlassos for reversible cargo capture, transport, and release.' " Langmuir. 2017 Jun 13;33(23):5932-7. https://doi.org/10.1021/acs.langmuir.7b00357 ; PubMed PMID: 28318267 . Cover http://pubs.acs.org/toc/langd5/33/23 , Jun-2017